Engine Model(s): PT6A

72-50-01

COMPRESSOR TURBINE STATOR - DESCRIPTION AND OPERATION

1. Description and Operation (Ref. 72-50-00, Fig. 1)

The compressor turbine classified guide vane ring consists of 29 vanes integrally cast between inner and outer rings. The airfoil section of the vanes directs the expanding gases from the combustion chamber to the compressor turbine blades at the optimum angle for maximum efficiency. The inner ring is cast with an extended rearward platform and internal flange. Lugs on the flange engage with slots in the No. 2 bearing cover flange.

An annular lockplate and 16 bolts secure the vane ring, No. 2 bearing cover and large exit duct to the gas generator case. The outer ring locates the compressor turbine shroud housing and small exit duct.

The compressor turbine shroud housing extends forward toward the power section. The extension provides two features: the outer diameter incorporates a circumferential groove which accommodates the interstage sealing ring(s) while the inner diameter is machined to accommodate the classified compressor turbine shroud segments. The segments are held in position by a retaining ring seated inside, and riveted to, the shroud housing. The shroud segments act as a seal and provide running clearance for the compressor turbine. The interstage sealing ring seals the mechanical separation between the combustion section and the interturbine area.

The large exit duct is located at the rear of the combustion chamber liner, adjacent to the diffuser in the front section of the gas generator case. The duct forms the outer section of an envelope to change direction of gas flow 180 degrees and directs it to the compressor turbine guide vanes. The duct, fabricated from nickel alloy, incorporates a heatshield at the rear. Compressor discharge air (P3) is routed into the space formed by the heatshield to cool the duct. The outer flange of the duct incorporates a "wiggle strip" Pre-SB1434 or sliding joint Post-SB1434 to provide a mating flange for the outer wall of the combustion chamber liner and ports for the passage of cooling air over the duct wall. The inner section of the duct incorporates a bolting flange for mounting at the gas generator case centerbore.

The small exit duct locates in the rear of the combustion chamber liner via a sliding joint and slotted seal ring and attaches to the compressor turbine guide vane ring. The duct forms the inner section of an envelope that changes the direction of gas flow 180 degrees and directs it to the compressor turbine inlet guide vanes. The duct is fabricated from nickel alloy and incorporates a heatshield and mounting flange at the front. Holes in the heatshield allow cooling air to be routed through the space between the duct wall and heatshield; the air then passes out of the space and into the gas stream at the ID of the duct wall.

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